Performance Testing

Date	12 July 2026
Team ID	AIML-HDI-2026
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the system behaves under expected and peak load conditions, and how accurate the trained model is. The sections below document the testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual browser testing of the Flask dashboard; scikit-learn metrics for model evaluation.
Type of Testing	Functional testing, model performance testing, and user acceptance testing.
Target Module / API	HDI prediction endpoint (Flask form submission → model inference → result render).
Test Environment	Kaggle Notebook GPU runtime, tunnelled publicly via LocalTunnel.
Test Date	12 July 2026

Step 2: Model Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Mean Absolute Error (MAE)	≤ 0.03	0.02640	Pass	Regression error well within target on held-out test split.
2	Root Mean Squared Error (RMSE)	≤ 0.04	0.03113	Pass	Consistent with the low MAE; no large outlier errors.
3	Tier Classification Accuracy	≥ 80%	83%	Pass	Verified on test set (n = 141) across 4 tiers.
4	Macro Avg. F1-score	≥ 0.75	0.83	Pass	“High” tier shows the most confusion with “Very High” near the boundary.

S.No	Metric	Target Value	Actual Value	Status	Remarks
5	Training Convergence	Stable by epoch 20	Stable ~epoch 10–20	Pass	Train/validation loss curves converge closely, indicating low overfitting.

Step 3: Functional & User Acceptance Testing

S.No	Test Case	Steps	Expected Result	Status
1	Web form loads correctly	Open the LocalTunnel dashboard URL in a browser.	Form renders with 3 numeric input fields and default values.	Pass
2	Valid input prediction	Enter Life Expectancy, Schooling, and GNI within valid ranges and submit.	Predicted HDI score and colour-coded tier badge display within seconds.	Pass
3	Boundary tier value	Enter values expected to produce a score near a tier boundary (e.g., ~0.70).	Correct tier (High vs Medium) is assigned per threshold rule.	Pass
4	Repeated submissions	Submit the form multiple times with different values in the same session.	Each submission returns an updated prediction without server restart.	Pass
5	Session stability	Keep the LocalTunnel session open for the duration of the demo.	Application remains reachable and responsive throughout.	Pass

Step 4: Observations & Analysis

Key Findings: The dual-branch hybrid model converges quickly (within the first 10–20 of 120 epochs) and generalizes well, with training and validation loss curves tracking closely throughout. The web dashboard returns predictions with no perceptible delay for a single user during manual testing.

Bottlenecks Identified: The primary limitation is the “High” tier's precision (0.63), where the regression output near tier boundaries is occasionally over-classified from adjacent tiers — an expected outcome of applying hard thresholds to a continuous prediction.

Optimization Steps Taken: Feature scaling (Min-Max normalization) was fit only on the training split and reused at inference time to avoid data leakage; the trained model and scaler were serialized separately so that inference does not require retraining.

Step 5: Screenshots / Evidence

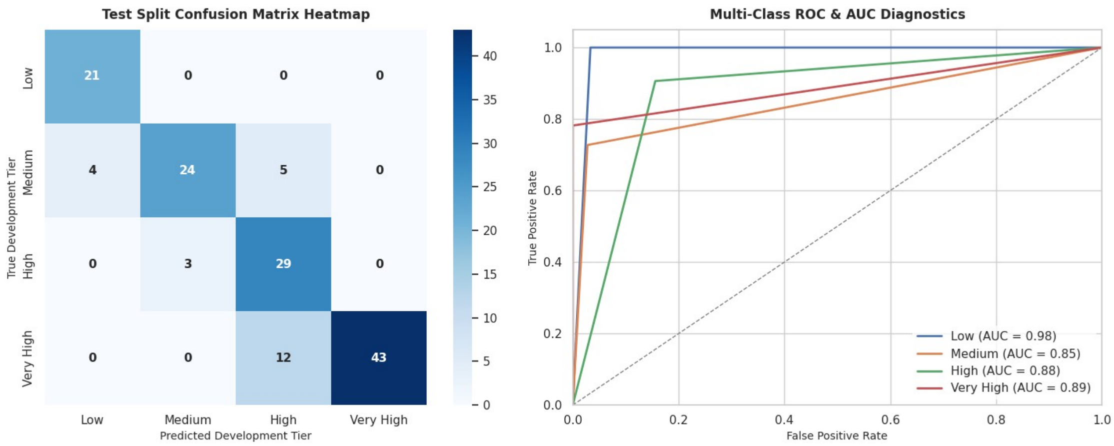

Figure: Test-set confusion matrix and multi-class ROC-AUC diagnostics, confirming the performance figures reported above.